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Energy Management of a Fuel Cell Electric Robot Based on Hydrogen Value and Battery Overcharge Control

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Abstract: The energy management system of a fuel cell electric robot should be highly responsive to provide the required power for various tactical operations, navigation of different routes, and acceleration. This paper presents a new multi-level online energy management strategy for a fuel cell electric robot based on the proposed functions of equivalent hydrogen fuel value evaluation, classification of the battery state of charge via the squared combined efficiency function, identification of the robot maneuver condition based on the proposed operation state of robot function, improvement of the overall energy efficiency based on the proposed function of the battery overcharge control, and separation of the functional points of the fuel cell based on the operational mode control strategy. The simulation study of the proposed online multi-level energy management strategy was carried out with MATLAB R2018b software to verify its superiority by comparing with other strategies. The results indicate a reduction in hydrogen consumption, reduction in fuel cell power fluctuations, prevention of battery overcharging, and incrementation in the total energy efficiency of energy storage systems compared to other energy management strategies.

Keywords: energy management strategy; battery; fuel cell; fuel value; combined efficiency; operation state; overcharge control



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1. Introduction

Nowadays, the field of electrified mobile robot design and construction is experiencing continuous growth and advancement. The combining of a power supply (fuel cell, photovoltaic, etc.) with an energy storage system (battery, supercapacitor, etc.) in a robot propulsion system represents a hybrid electric robot concept [1]. This hybridization offers potential solutions to overcome the challenges associated with pure electric robots. In the first case, if only batteries are used as the energy source, they will encounter numerous obstacles including low power density, voltage drop, and inefficiency performance. Also, if only fuel cells are used as the energy source, the fluctuations of the demanded power can reduce the life span of the fuel cell. Consequently, the fuel cell electric robot is usually equipped with batteries or supercapacitors as an auxiliary energy source [2,3].

This approach has the potential to significantly enhance energy efficiency and prolong the operational capabilities of electric robots in hard operating conditions [4]. Meantime, the energy management and power control strategy of hybrid power sources are crucial.

The energy management strategy necessitates a comprehensive approach to addressing the challenges in the domains of power sources, maneuverability, operational durability, lifespan, and efficiency [5]. Fuel cell electric robots often face various emergencies based on path planning; the maneuver conditions of mobile robots are very complex, and the demanded power will also have large fluctuations and sudden changes [6,7].

The power allocation system in a robot with hybrid power sources should be optimized to efficiently meet the power demands of various tactical operations, navigation across

different routes, and acceleration. Furthermore, considering the importance of operation duration in these robots, it is crucial to implement a power and energy management strategy that not only optimally distributes power among the hybrid power sources, but also minimizes hydrogen consumption and maximizes power system efficiency [4,8]. In addition, enhancing the robot's ability to recycle and utilize energy, prolonging battery lifespan, minimizing fluctuations in the initial and final state of charge (SOC) of the battery, and reducing fluctuations in the fuel cell output power are some of the factors that can be incorporated into the multi-objective optimization function [9].

To enhance the agility of mobile robots in various directions, it is necessary to have various power sources with distinct performance metrics. While the battery is a crucial source of power generation, it is insufficient to meet the power demands of dynamic loads, particularly fast dynamic loads. Additionally, charging the battery under various conditions can pose challenges. Consequently, incorporating alternative power sources like fuel cells and supercapacitors can enhance the efficiency of mobile robots [10,11].

Once power supply sources are selected, various approaches to achieve optimal power distribution in mobile robots have become crucial for effective power and energy management strategies. Implementing an optimal energy management strategy (EMS) on a mobile robot can result in enhanced maneuverability, improved energy efficiency, and increased reliability.

The primary objective of the energy management strategy is to accurately allocate the power of the fuel cell and the battery in real-time, with the aim of achieving various goals such as minimizing hydrogen fuel consumption, regulating battery overcharging, minimizing fluctuations in the fuel cell output power, maintaining the permissible range of the battery state of charge, optimizing system efficiency, and so on.

This article is organized into six sections. The second part presents an overview of the current EMS strategies and challenges in the field. The third part of the course introduces the fundamental principles and techniques involved in the theory and modeling of power supply sources, specifically focusing on fuel cells and batteries. The fourth part presents an integrated strategy that encompasses several key components. These include a state machine control strategy that is based on the robot's state machine function, a proposed function that is based on the equivalent consumption minimization strategy, an operating mode control strategy that is based on the valuation of the consumed hydrogen, and a battery overcharge control function. The fifth part presents the modeling of the proposed strategy using MATLAB R2018b software and the subsequent analysis of its results. The sixth section entails a thorough comparison and discussion of the results.

2. Literature Review

An online energy management strategy should be implemented in electric robots. Offline methods necessitate prior knowledge of the robot's maneuvering conditions, pre-mission power requirements, mission duration, and other parameters that cannot be anticipated or determined in real-time conditions [12].

Online energy management can be categorized into two groups: rule-based and optimization-based strategies. The participation rate of each power supply source in rule-based management is determined by decision-making rules, which can be established by research centers, mathematical models, experts, or innovative methods [13]. Various decision-making methods based on computational rules exist, including the state machine control strategy (SMCS), power flow control strategy, modified power flow strategy, thermostat strategy, operational mode control strategy (OMCS), and strength coefficient mode control strategy [14].

Optimization-based energy management methods can be optimized using various approaches, such as equivalent consumption minimization strategy (ECMS), predictive control model, robust control methods, intelligent control, and more. ECMS, among the optimization-based methods, is capable of identifying the optimal estimation point in a shorter time frame [10]. In recent years, the utilization of both optimization-based and rule-based categories has been increased in order to implement integrated energy management strategies [15].

Lü et al. [3] proposes a combined strategy that utilizes state machine control and fuzzy logic to enhance the performance of a power supply system that comprises a proton exchange membrane fuel cell (PEMFC) and a battery used in a mobile electric robot. The findings from Reference [3] demonstrate a decrease in hydrogen fuel consumption and variations in the power generation of PEMFC fuel cells. However, the investigations have neglected to consider the changes in the battery state of charge mentioned in this reference.

The power management of a mobile electric robot, equipped with a proton exchange membrane fuel cell and battery, is achieved by employing a power-split control strategy and fuel cell air circulation management, as described by Figueroa et al. [16]. In the cited source, two PI controllers are employed to regulate the air circulation and electric current of the fuel cell. The controller's adjustment is based on the ideal reference points of the robot's average power consumption and operation duration. The aforementioned study has solely focused on enhancing the functional aspects of the fuel cell while disregarding any efforts to enhance the functional aspects of the battery.

The article presented by Boukoberine et al. [17] discusses the management of energy in an unmanned aerial vehicle (UAV) that is equipped with both a fuel cell and a battery. The energy management strategy described in the reference utilizes a combination of rule-based methods and an equivalent fuel consumption minimization strategy. This approach is implemented to enhance power supply efficiency and reduce hydrogen consumption. The implementation of the strategy described in the mentioned reference on a quad-copter with an average power of 650 watts resulted in a 3% reduction in hydrogen fuel consumption. This reduction has the potential to lead to a 21.8-min increase in the robot's maneuver duration. The separation of rule-based conditions in the mentioned reference is based on engineering experience rather than valid mathematical relations.

Zhou et al. [18] introduces a novel quadratic sliding mode control technique designed to minimize fuel cell usage in a hybrid power system comprising a supercapacitor and a fuel cell. This method involves the utilization of a robust non-linear cascade control loop to effectively regulate the PEMFC fuel cell. The primary objective is to minimize the fluctuations in both the electric current and output power of the PEMFC. Nevertheless, the presented method is not capable of real-time execution and operates offline. Additionally, although the installation of a supercapacitor has reduced the current ripple in the fuel cell output, there are no available data regarding the specific extent of the reduction or any potential impact on fuel consumption.

In recent years, the study of energy management strategies has led to improved stability and efficiency in power supplies. However, the development of EMS power supplies remains a crucial challenge in fuel cell electric robots. The previous EMS presentation did not adequately address the functional, economic, and security limitations of power supply sources, specifically fuel cells and batteries. The studies primarily focused on power allocation between fuel cells and batteries, neglecting other important issues such as stresses on the fuel cell, the lifespan of the fuel cell, and the operational efficiency of power supply sources such as batteries. Hence, enhancing power control and implementing comprehensive energy management is regarded as a fundamental obstacle in enhancing the performance of fuel cell electric robots.

To address the challenges mentioned, a novel multi-mode online energy management system is proposed. This system incorporates state machine control (SMC), operational mode control (OMC), and an equivalent consumption minimization strategy (ECMS) for efficient power and energy management, as depicted in Figure 1. The definition of the proposed EMS is based on binary code which can only be one or zero in each real condition ($d = 0$, or 1). Different levels of the proposed EMS are converted to binary codes, and a binary string of bits ($[d \ d]$ or $[d \ d \ d]$) is used to specify operation modes. Therefore, a new online EMS is presented with the following contributions.

- A novel state machine function has been established for a fuel cell electric robot that is equipped with a battery, enabling it to discern various states of the robot's maneuver.

- The reduction in power fluctuations in the fuel cell output has been successfully implemented through the proposed strategy.
- Novel mathematical equations have been developed to express the relationship between the battery formulation and the battery SOC. These equations are derived through statistical analysis, in contrast to previous studies that relied on experimental methods.
- Novel mathematical equations have been suggested to calculate the equivalent hydrogen consumption fuel value (EHCFV) as part of the implementation of the equivalent fuel consumption minimization strategy.
- This project introduces a novel concept called battery overcharge control (BOCC) to effectively prevent the unnecessary and excessive charging of batteries.

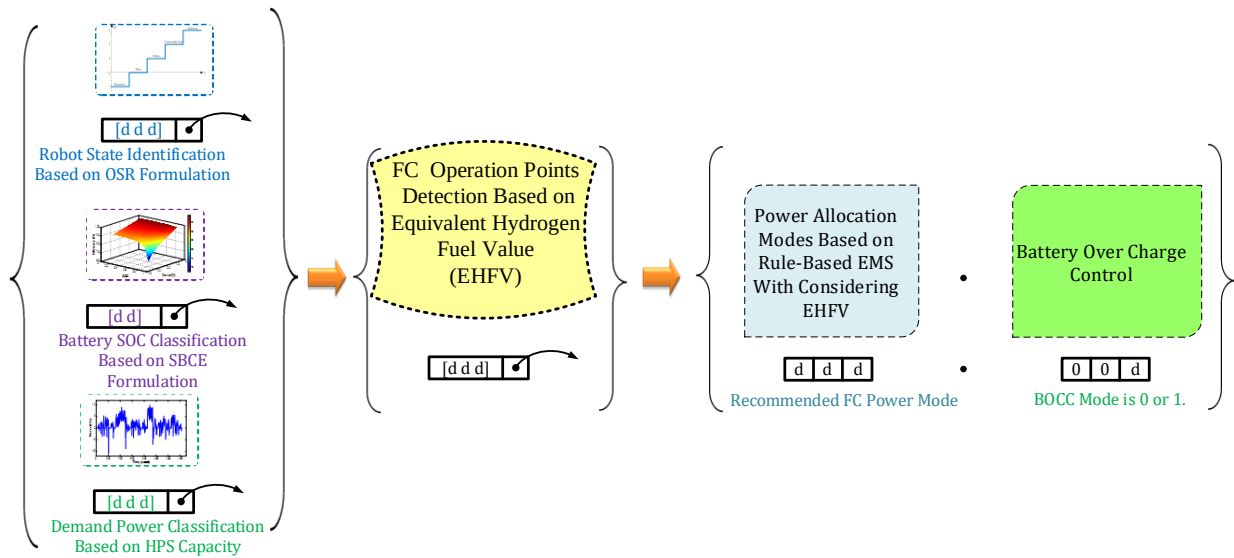


Figure 1. Schematic diagram of the fuel cell electric robot and proposed energy management strategy.

3. Materials and Methods

The functionality of each individual component of the robot is crucial, and its dynamic equations and analysis enable us to assess the performance of various parts of the robot and its energy management strategy with ease. This section aims to establish the scientific foundations for evaluating the energy management strategy of a fuel cell electric robot (including PEMFC and a LiFePO₄ battery). This will be performed by utilizing electrochemical equations of the power supply sources and static equations of the electronic power converters. The research focuses on selecting the fuel cell electric robot with a battery and PEMFC as the target structure for the hybrid power source, as shown in Figure 2.

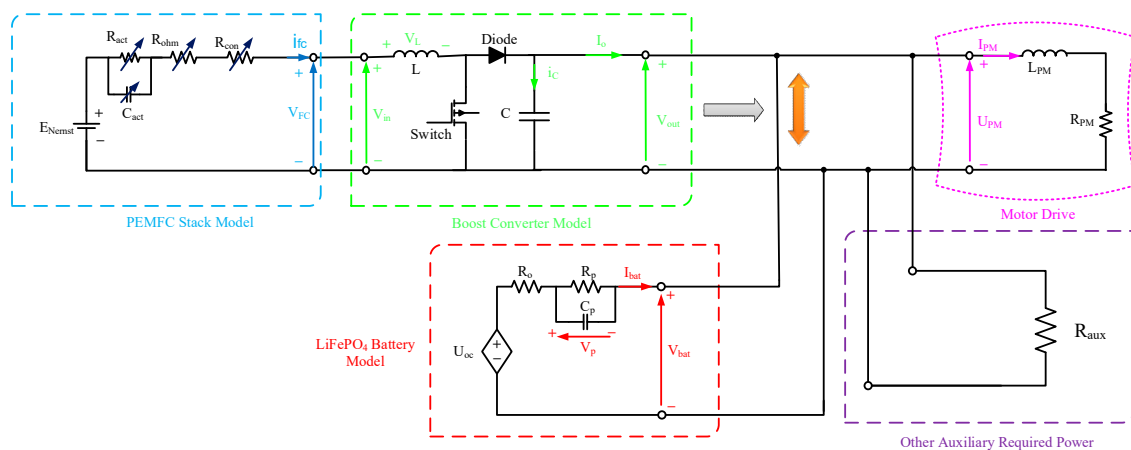


Figure 2. The equivalent circuit of the target structure of the power chain in the robot.

3.1. Fuel Cell Modeling

It is crucial to identify, model, and optimize the control of the fuel cell in order to effectively manage the power of the fuel cell electric robot. The typical power supply system for a PEMFC includes a fuel cell stack and various secondary components, such as a hydrogen tank, condenser, water pump, air compressor, water tank, humidifier, DC-DC converter, and water radiator. The electrical equivalent circuit of a PEMFC is depicted in Figure 2, illustrating its dynamic model. In this circuit, E_{Nernst} represents the voltage generated by the chemical reaction, R_{act} represents the resistance associated with the activation of the reaction, R_{con} represents the resistance caused by the saturation of oxygen gas concentration, R_{ohm} represents the resistance related to ohmic losses, and C_{act} is the equivalent reaction activation capacitor.

The power of the fuel cell (P_{FC}), as described by Equation (1), is determined by multiplying the FC voltage (V_{FC}) and current of the fuel cell (i_{FC}). The efficiency of the stack and ventilation system of the fuel cell (η_{FC}) is defined by Equation (2). In this equation, P_{aux} represents the power consumed by the secondary components, P_{H2} represents the power generated by hydrogen (as stated in Equation (3)), m_{H2} represents the rate of hydrogen consumption (as stated in Equation (4)), ΔH represents the change in enthalpy of hydrogen, and ρ_{H2} represents the chemical energy density of the moles of hydrogen [19].

$$P_{FC} = V_{FC} i_{FC} \quad (1)$$

$$\eta_{FC} = \frac{P_{FC} - P_{aux}}{P_{H2}} \quad (2)$$

$$P_{H2} = m_{H2} \Delta H \quad (3)$$

$$m_{H2} = \int \frac{P_{FC}(t)}{\eta_{FC} \times \rho_{H2}} dt \quad (4)$$

While the fuel cell stack is essential to the system, secondary components such as an air compressor, humidifier, pressure regulator, and fluid circulation rate also play a crucial role in its operation. Additionally, a portion of the fuel cell production power is consumed by these components. Hence, it is imperative to incorporate the internal power consumption of the auxiliary components (P_{aux}) into the efficiency calculation of the fuel cell system. The secondary components of a PEMFC system can be categorized into four subgroups: hydrogen circulation circuit, air circulation circuit, humidifier circuit, and cooling circuit. The internal power consumption of a PEMFC is determined by the combined power consumption of its four subgroups, as represented by Equation (5) [20].

$$P_{aux}(t) = P_0 + P_{hp}(t) + P_{em}(t) + P_{ahp}(t) + P_{cl}(t) + P_{cf}(t) \quad (5)$$

where P_0 represents the power generated by the bias current of the fuel cell. The bias current corresponds to the minimum current necessary to sustain the reaction and maintain the operational state of the fuel cell. The power required for the hydrogen circulation circuit (P_{hp}) is estimated using Equation (6), where K_{hp} represents the constant coefficient associated with the circuit. The power consumption necessary for air circulation by the compressor (P_{em}) has been approximated using Equation (7), where K_{cp} represents the constant coefficient associated with air circulation.

$$P_{hp}(t) = K_{hp} \times i_{FC}^2(t) \quad (6)$$

$$P_{em}(t) = K_{cp} \times i_{FC}(t) \quad (7)$$

The power consumption of the humidifier system (P_{ahp}) can be approximated by using Equation (8), in which K_{ahp} represents the constant coefficient associated with air circulation.

$$P_{ahp}(t) = K_{ahp} \times i_{FC}(t) \quad (8)$$

The power consumption of the cooling system is possible to express by two distinct circuits: air cooling (P_{cf}) and water cooling (P_{cl}). This relationship can be expressed using Equation (9):

$$P_{cl}(t) + P_{cf}(t) = K_{cl1} \times i_{FC}(t) + K_{cl2} \times i_{FC}^2(t) \quad (9)$$

where K_{cl1} and K_{cl2} represent fixed coefficients for air and water cooling, respectively. Equations (6)–(9) provide a basis for approximating the internal power consumption of a PEMFC system, as shown in Equation (10). This approximation involves the use of the constant coefficients K_1 (coefficient of voltage with unit measurement of volt) and K_2 (coefficient of resistance with unit measurement of ohm) [16].

$$P_{aux}(t) = P_0 + K_1 \times i_{FC}(t) + K_2 \times i_{FC}^2(t) \quad (10)$$

3.2. Modeling of Non-Isolated Step-Up Converter

The fuel cell electric robot relies on a fuel cell as its power source for various tasks such as movement, acceleration, and maneuvering in different conditions. To accomplish this, it is imperative to provide the robot with the necessary power to traverse various routes at the correct voltage. The optimal voltage for the DC link in a robot typically exceeds the output voltage of the PEMFC stack. Thus, it appears imperative to employ a step-up converter in power generation sources.

It is important to acknowledge that the fuel cell's output voltage is variable, and there is a significant disparity in voltage between no-load and full-load modes. The voltage of a single-cell PEMFC in full load mode is 0.6 V, whereas its no-load voltage is 1.23 V. An instance of this can be seen in the Horizon fuel cell, which has an output power of 500 watts and utilizes 24 cells. In full-load mode ($V_{FC, Full Load}$), the voltage of this fuel cell is 14 volts, whereas in no-load mode ($V_{FC, No Load}$), the voltage is approximately 22.5 volts. Consequently, fuel cells require a DC-DC converter to modify the output voltage according to various applications. The static gain (M) and efficiency (η_{dc}) of the non-isolated step-up converter can be determined using the volt-second balance law and the charge current balance law in continuous operating mode. These calculations are based on Equations (11) and (12), where R_L represents the resistance resulting from inductance, R_{on} represents the switch resistance during conduction, R_D represents the diode resistance during conduction, and V_D represents the diode conduction voltage [21,22].

$$M = \left(\frac{1}{1-D} \right) \left(1 - \frac{(1-D)V_D}{V_s} \right) \left(\frac{(1-D)^2 R}{(1-D)^2 R + R_L + DR_{on} + (1-D)R_D} \right) \quad (11)$$

$$\eta_{dc} = \left(\frac{\left(1 - \frac{(1-D)V_D}{V_s} \right)}{1 + \frac{R_L + DR_{on} + (1-D)R_D}{(1-D)^2 R}} \right) \quad (12)$$

3.3. Battery Modeling

The LiFePO₄ battery, also known as LFP, is the optimal choice for fuel cell electric robots due to its unique power capabilities, extended lifespan, and high energy efficiency. A precise battery model is necessary to ensure the exact simulation of battery performance. The battery's dynamic model is contingent upon the internal parameters and state of charge, and a thorough analysis of it is crucial for effectively managing the power of hybrid power sources. The LiFePO₄ battery model's framework is depicted in Figure 2, with U_{oc} representing the open circuit voltage of the LFP battery. The value of U_{oc} varies depending on the state of charge or discharge. C_p represents the capacitor that is equivalent to the LFP battery, while R_p represents the parallel resistance that is connected to the capacitor and is equivalent to the potential difference V_p . The R_o resistance refers to the total resistance in the equivalent circuit, while I_{bat} represents the current flowing through the battery and V_{bat} represents the voltage output of the battery. The regression method was used to identify each of the LFP battery modeling parameters [23].

Once the battery modeling parameters have been identified, the SOC of the battery can be calculated using Equation (14) [24].

$$\begin{cases} V_{\text{bat}} = U_{\text{oc}} - R_{\text{O}}I_{\text{bat}} - V_{\text{p}} \\ \dot{V}_{\text{p}} = -(1/R_{\text{p}}C_{\text{p}})V_{\text{p}} + (1/C_{\text{p}})I_{\text{bat}} \end{cases} \quad (13)$$

$$\text{SOC} = \text{SOC}_0 - \frac{\int_0^t I_{\text{bat}} dt}{Q_e} \quad (14)$$

The initial SOC of the battery is denoted as SOC_0 , and Q_e represents the battery capacity. The open circuit voltage (U_{oc}) and internal resistances of the battery are influenced by the SOC of the battery. This, in turn, affects the charging and discharging efficiency of the battery in a steady state, as described by Equations (15) and (16), as follows:

$$\eta_{\text{B,chg}}(\text{SOC}, P_{\text{B}}) = \frac{2}{1 + \sqrt{1 + \frac{4(R_{\text{o}} + R_{\text{p}})P_{\text{B}}}{U_{\text{oc,chg}}^2}}} \quad (15)$$

$$\eta_{\text{B,dis}}(\text{SOC}, P_{\text{B}}) = \frac{1 + \sqrt{1 - \frac{4(R_{\text{o}} + R_{\text{p}})P_{\text{B}}}{U_{\text{oc,dis}}^2}}}{2} \quad (16)$$

4. The Proposed Energy Management Strategy

The research findings indicate that power and energy management strategies in the domain of electric robots face several limitations. The aforementioned strategies typically prioritize the distribution of power from each source (fuel cell and battery) in order to minimize fuel consumption. However, they do not address concerns related to power generation fluctuations in the fuel cell, optimizing battery charge and discharge, and achieving the desired final charge level. The battery and the stress on the fuel cell have not been inspected. Hence, offering an efficient proposed approach to power regulation and energy management can be regarded as a crucial aspect to addressing the previously mentioned issues.

The energy management strategy proposed in this research combines equivalent consumption minimization strategies, operational mode control, state machine control, and battery overcharge control. This strategy involves managing operational states and modes according to analysis and mathematical correlations. The proposed energy management strategy in the fuel cell electric robot involves a novel approach to identify machine states, determine the SOC limits of the LFP battery using the SBCE concept, introduce the equivalent value of hydrogen consumed during battery charging and discharging (measured in grams per joule), and control battery overcharge. Figure 3 illustrates the utilization of established concepts in a fuel cell that is equipped with a battery.

4.1. Robot Maneuver State Identification Strategy

Currently, in various energy management strategies, fluctuations in the power production of fuel cells and batteries are not effectively mitigated, and the stages of identifying different maneuver conditions are not carried out. To address the aforementioned difficulties, a novel operation state of robot (OSR) function has been established. This function utilizes the robot's acceleration, movement speed, and power requirements to identify various states of robot movement. The newly introduced state function will be utilized to execute the state machine control strategy. Furthermore, once the OSR function is defined, the state machine control strategy will be employed to minimize the initial variations in the production power of PEMFC. Additionally, this strategy will also consolidate the operational modes of the fuel cell and battery.

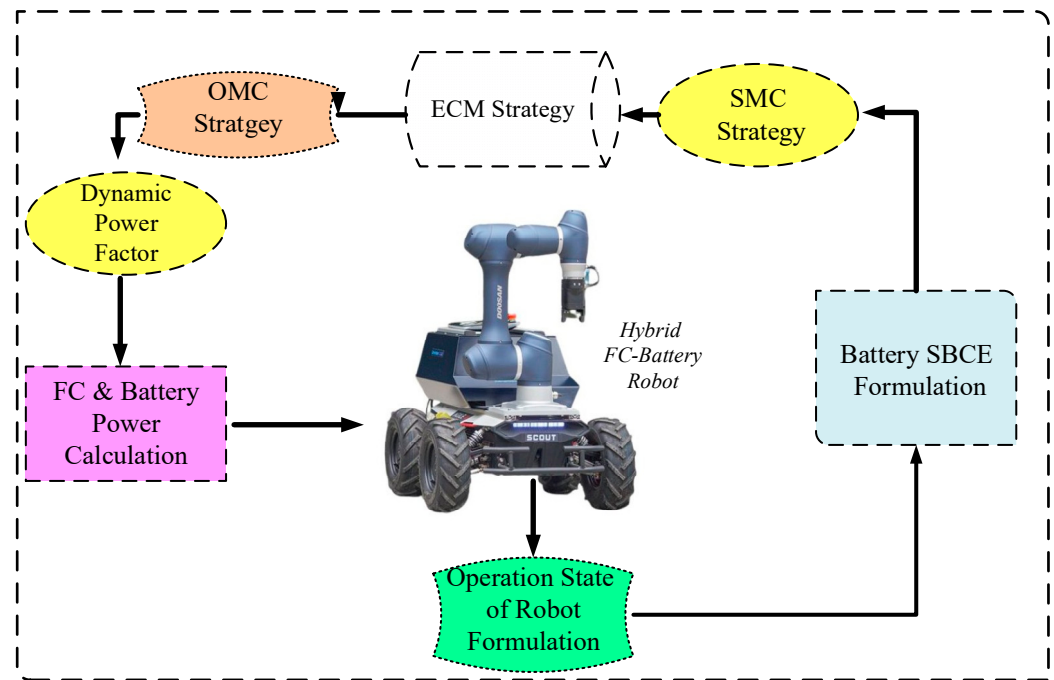


Figure 3. The utilization of proposed energy management strategy concept in a fuel cell electric robot.

Thus far, there has been no report in the field of robotics that specifically addresses the identification of a robot's movement state. However, previous literature in the automotive field commonly categorizes the states of an automobile's movement as stop/brake/acceleration. The proposed formulation aims to utilize various values of power demand (P_{demand}), changes in power demand (ΔP_{demand}), and changes in robot movement speed (Δv) as the fundamental concept for determining different states of the robot's status. Equation (17) presents the fundamental idea of the OSR and its formulation.

$$OSR(t) = \text{sign}(P_{demand}(t)) + \text{sign}(\Delta P_{demand}) + \text{sign}(\Delta v) \quad (17)$$

In the above, the 5 different operation states of the robot for different conditions are as follows:

- Stationary state: $OSR = 0$;
- Acceleration state: $OSR = 3$;
- Traction state: $OSR = 2$;
- Deceleration mode: $OSR = 1$;
- Regenerative braking state: $OSR = -1$.

4.2. Battery SOC Zone Classification Based on the Proposed Function

This paper introduces the concept of a square of battery combined energy (SBCE), which allows for the utilization of mathematical relationships to distinguish the zone of battery SOC. Furthermore, it emphasizes the significance of battery efficiency in power and energy management equations.

The research defines the square of battery combined energy (η_{SBCE}) by considering the average efficiency of both charging and discharging the battery, as described in Equation (18).

$$\eta_{SBCE} = \left(\frac{\eta_{B,dis} + \eta_{B,chg}}{2} \right)^2 = \left(\frac{1 + \sqrt{1 - \frac{4(R_o + R_p)P_B}{U_{oc,dis}^2}}}{4} + \frac{1}{1 + \sqrt{1 + \frac{4(R_o + R_p)P_B}{U_{oc,chg}^2}}} \right)^2 \quad (18)$$

Figure 4 presents the η_{SBCE} function for various battery powers and SOC in per-unit status, addressing multiple issues simultaneously. Given the inherent difficulty in accurately determining the charging or discharging status of the battery while it is operating on the fuel cell electric robot, as well as the varying efficiency of the battery during charging and discharging, Equation (18) provides a clear definition for the process of charging the battery at one moment and discharging it in subsequent moments (or vice versa).

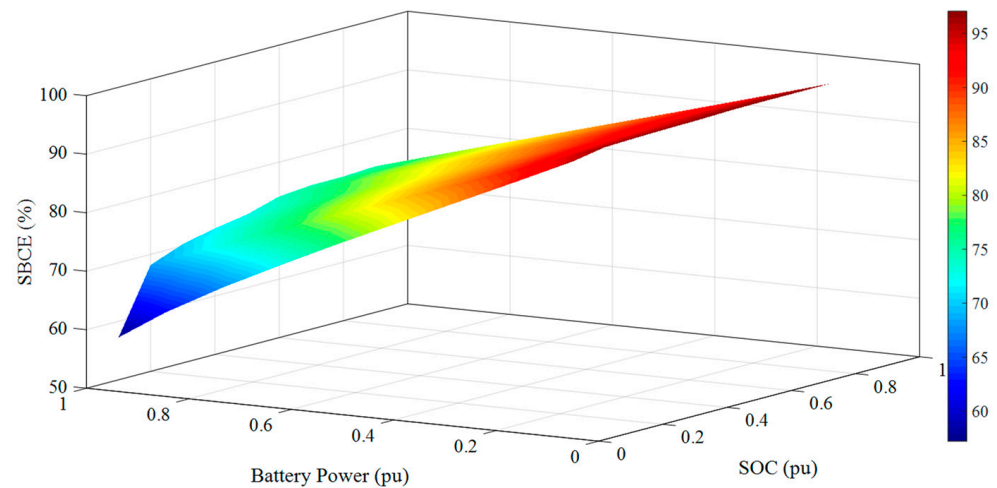


Figure 4. The LFP battery squared combined efficiency curve in per-unit status.

In the given relationship, all the variables (excluding battery power) rely on the battery charge level. Hence, in order to distinguish between various battery charge states, the nominal battery power will be utilized to determine the sensitivity of SBCE to SOC.

This study utilizes the concepts of mathematical expectation and standard deviation to examine the sensitivity of SBCE to SOC. In order to achieve this goal, we define the mathematical expectation of the SBCE (μ_{SBCE}) using Equation (19).

$$\mu_{SBCE} = \sum_{i=1}^n \eta_{SBCE,i} P_{SBCE}(\eta_{SBCE,i}) \quad (19)$$

In this equation, $\eta_{SBCE,i}$ represents the SBCE for various SOC, and $P_{SBCE}(\eta_{SBCE,i})$ represents the corresponding probability distribution function for different SOC of the LiFePO₄ battery. This PDF follows a uniform distribution in the range of (0.1, 0.9) pu for the SOC. In this study, the dispersion of the SBCE and the SOC range were analyzed using the dispersion index of the standard deviation, after calculating μ_{SBCE} .

The mentioned dispersion index quantifies the extent to which SBCE deviates from the mean point. The standard deviation (σ_{SBCE}) is calculated by taking the square root of the variance of the SBCE, as given by Equation (20).

$$\sigma_{SBCE} = \sqrt{E((SBCE - \mu_{SBCE})^2)} = \sqrt{\sum_{i=1}^n (\eta_{SBCE,i} - \mu_{SBCE})^2 P_{SBCE}(\eta_{SBCE,i})} \quad (20)$$

Figure 5 displays the SBCE of the LFP battery, along with its mean value and standard deviation, for various SOC of a LiFePO₄ battery cell operating at its rated power. The maximum SBCE is 75%, achieved at SOC = 0.9 pu. The lowest SBCE is 57%, observed at SOC = 0.2 pu. μ_{SBCE} is 72.44% and its standard deviation (σ_{SBCE}) is 5.6%. These data were utilized to determine the functional areas of the battery. The division of the battery's 4 functional areas based on their maximum values, mean, and standard deviation is outlined in Figures 5 and 6, represented as binary codes.

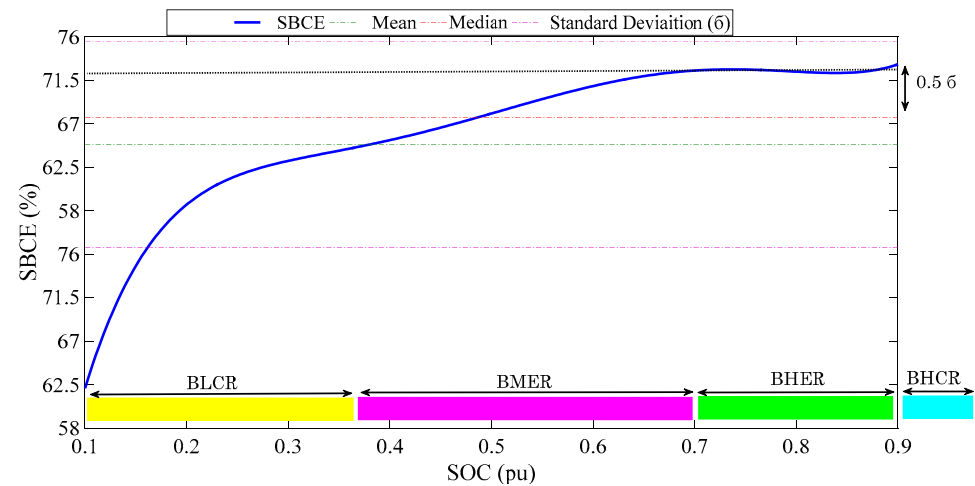


Figure 5. The SBCE, dispersion, and deviation of the LFP battery SOC zone based on the proposed.

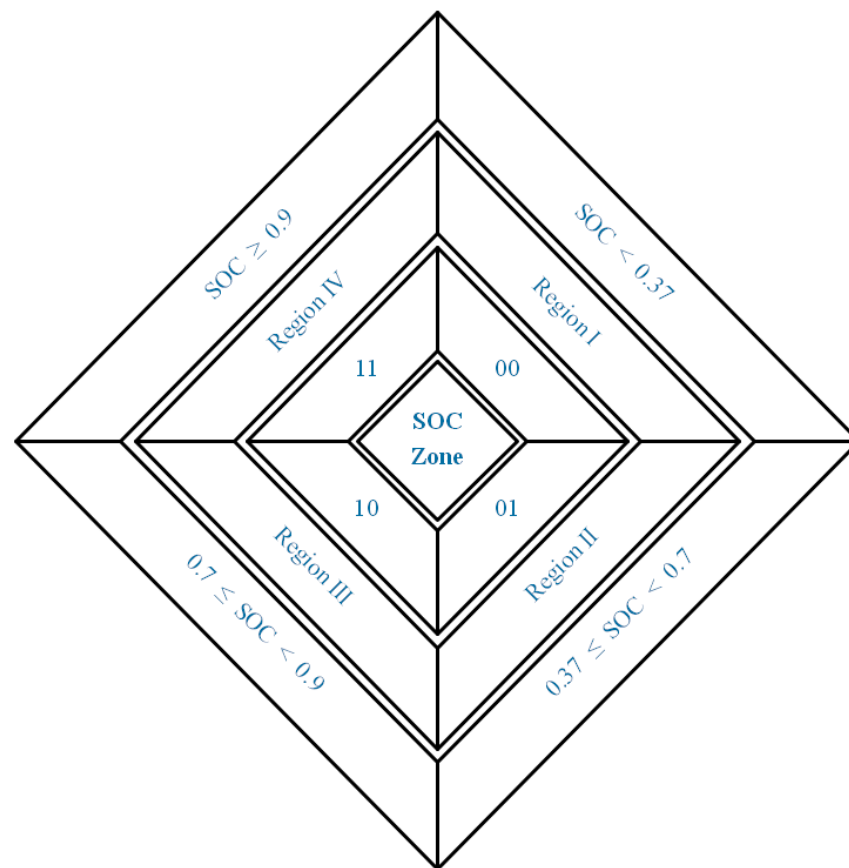


Figure 6. Detected binary codes of LFP battery SOC classification in proposed EMS function.

Based on the cases mentioned, the average SBCE of the battery is equal to its value at $SOC = 0.37$ pu, which is selected as the initial boundary point. Based on the figure, the SOC value of 0.7 pu represents the point at which the SBCE of the $LiFePO_4$ battery has stabilized. At this SOC, the efficiency is equal to the average value of the SBCE and half of its standard deviation ($0.5 \sigma_{SBCE} + \mu_{HC}$). Thus, the point $SOC = 0.7$ pu has been selected as the second boundary point. Once the SOC of the battery reaches 0.9, the battery's capacity for charging is depleted. As a result, $SOC = 0.9$ pu has been selected as the third boundary point.

4.3. Formulating Hydrogen Fuel Valuation

Figure 1 illustrates the overall configuration of the power supply system for the fuel cell electric robot, which includes a battery. The power supply for the robot is provided by a PEM type fuel cell and a LiFePO₄ type battery in this configuration. Based on the diagram, the total efficiency of the fuel cell system, including the DC-DC step-up converter, is determined by multiplying the efficiencies of the fuel cell and the converter. This relationship is expressed by Equation (21).

$$\eta_{\text{tot}} = \eta_{\text{FC}} \times \eta_{\text{dc}} \quad (21)$$

The efficiency of the DC-DC step-up converter is denoted as η_{dc} , while the efficiency of the fuel cell is denoted as η_{FC} , as defined in the previous sections. In addition, after computing η_{tot} , the battery's equivalent fuel consumption is calculated. The battery's equivalent fuel consumption is determined by factors such as SOC changes, battery capacity, battery efficiency, and the power electronic equipment in its circuit. Hence, the ultimate expression for determining the equivalent fuel consumption is derived from Equation (22).

$$m_{\text{equivalent,H2}} = \frac{m_{\text{H2}}}{\eta_{\text{tot}}} + \text{HCJE}_{\text{bat}} = \frac{m_{\text{H2}}}{\eta_{\text{tot}}} + \frac{\Delta \text{SoC} \times E_{\text{batt}} \times 3600}{\eta_{\text{tot}} \eta_{\text{SBCE}} \rho_{\text{H2}}} \quad (22)$$

The variable, HCJE_{bat} , represents the amount of hydrogen consumed per joule of energy during battery charging, while ρ_{H2} represents the chemical energy density of hydrogen in megajoules per kilogram.

The purpose of this study is to establish the value of each gram of hydrogen fuel consumed for generating electric energy. This will be achieved by defining the rate of hydrogen consumption per joule of energy, denoted as m_{H2} , using Equation (23).

$$\text{HCJE}_{\text{H2}} = \frac{m_{\text{H2}}}{P_{\text{FC}} \eta_{\text{tot}}} \quad (23)$$

By substituting the appropriate equations, Equation (24) yields the rate of hydrogen consumption per joule of energy, denoted as HCJE_{H2} .

$$\text{HCJE}_{\text{H2}} = \frac{\left(\int_0^t \frac{M_{\text{H2N}}}{2F} i_{\text{FC}}(t) dt \right)^2 \times \Delta H}{\left(\frac{\left(1 - \frac{(1-D)V_D}{V_s} \right)}{1 + \frac{R_L + DR_{\text{on}} + (1-D)R_D}{(1-D)^2 \frac{V_Q^2}{V_{\text{FC}}^2 i_{\text{FC}}}} \right) (V_{\text{FC}} i_{\text{FC}} - P_{\text{aux}}) V_{\text{FC}} i_{\text{FC}}} \quad (24)$$

In the given relationship, all the parameters (excluding the fixed ohmic values in the converter) rely on the current-voltage (power) of the fuel cell output. Consequently, in order to distinguish the most efficient performance points of the fuel cell in relation to equivalent fuel consumption, the power variations in the fuel cell will be used to calculate the HCJE_{H2} sensitivity.

This study employs the concepts of mathematical expectation and standard deviation to examine the sensitivity of HCJE_{H2} to fuel cell power. Equation (25) defines the mathematical expectation of the hydrogen consumption rate per joule of energy (μ_{HC}).

$$\mu_{\text{HC}} = E(\text{HC}) = \int_0^{P_{\text{FC}}} \text{HCJE}_{\text{H2}i} P_{\text{H}}(\text{HCJE}_{\text{H2}i}) d(P_{\text{FC}}) \quad (25)$$

The variable $\text{HCJE}_{\text{H2}i}$ represents the rate at which hydrogen is consumed per joule of energy under various power conditions. The variable $P_{\text{H}}(\text{HCJE}_{\text{H2}i})$ represents the probability distribution function associated with uniform distribution in the per-unit interval (1, 0). In this study, after computing μ_{HC} , the aim is to examine the distribution of the hydrogen consumption rate per joule of energy and separation. The variability of the standard deviation index is utilized to ascertain the operational thresholds of the fuel cell.

The dispersion index mentioned quantifies the variability of hydrogen consumption per joule of energy from the midpoint. The standard deviation (σ_{HC}) is calculated using Equation (26) by taking the square root of the variance of $HCJE_{H_2}$.

$$\sigma_{HC} = \sqrt{E((HC - \mu_{HC})^2)} = \sqrt{\int_0^{P_{FC}} (HCJE_{H_2i} - \mu_{HC})^2 P_H(HCJE_{H_2i}) d(P_{FC})} \quad (26)$$

The forthcoming section will present the values of hydrogen consumption rate per joule of energy, including the mean value (μ_{HC}) and its standard deviation (σ_{HC}), for various powers. Subsequently, the fuel cell's various functional areas will be determined by utilizing the minimum values of $HCJE_{H_2}$, μ_{HC} , and σ_{HC} . This determination is crucial for implementing energy management strategies such as ECMS and OMC.

4.4. Fuel Cell Operational Modes

Figure 7 displays the efficiency–duty cycle curves and voltage gain–duty cycle curves for a modular structure of the DC-DC step-up converter on a per-unit basis, as determined by the equations discussed in the previous sections of this research. Figure 8 displays the overall efficiency (η_{tot}) of the fuel cell system, which is determined by multiplying the efficiency of the fuel cell and the step-up converter connected to it, after calculating the efficiency of the stack and the converter. The figure indicates that the system's efficiency is significantly lower than expected for output powers below 0.17 pu. To optimize its usage, this research demonstrates the appropriate operating range for PEMFC (PFC $\in (0.17, 1)$ pu) in Figure 8.

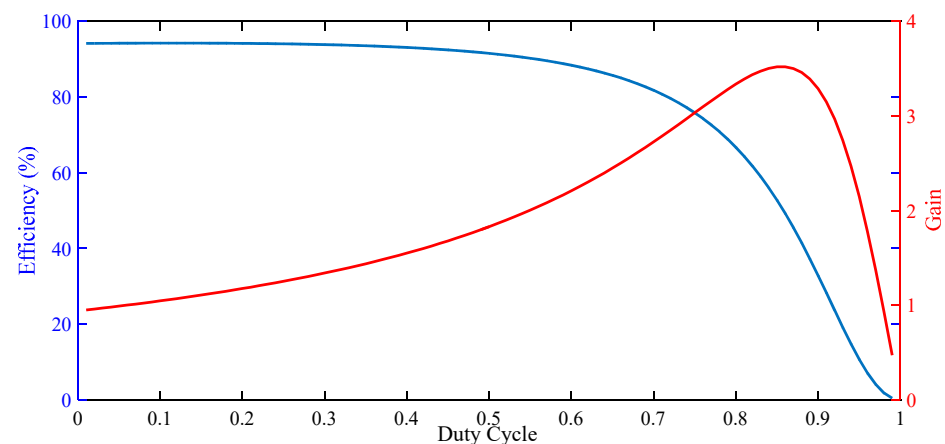


Figure 7. Efficiency–duty cycle and gain–duty cycle curves of the step-up converter.

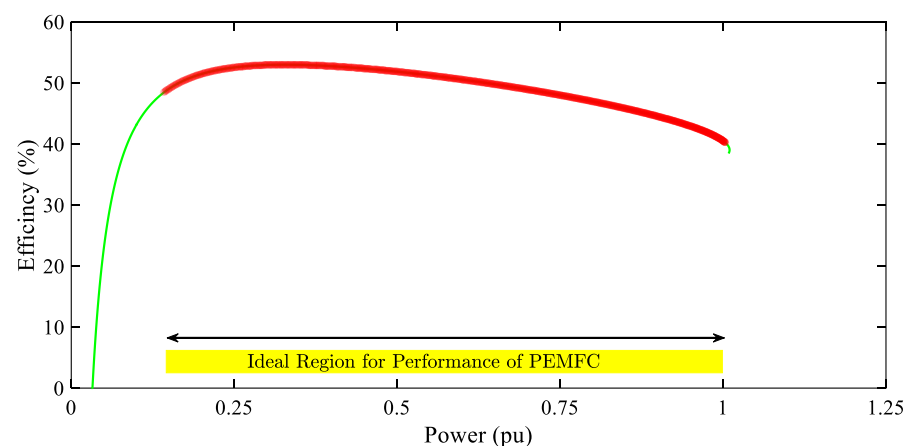


Figure 8. The total efficiency of the PEMFC system equipped with a linked step-up converter.

Figure 9 shows the simulation results for the rate of hydrogen consumption per joule of energy (HCJE_{H_2}), the mean value (μ_{HC}), and its standard deviation (σ_{HC}) in different operating conditions of the PEMFC modular structure. Figure 10 provides binary characteristic codes that specify the separation of the six operating points of the fuel cell based on the minimum values of HCJE_{H_2} , μ_{HC} , and σ_{HC} .

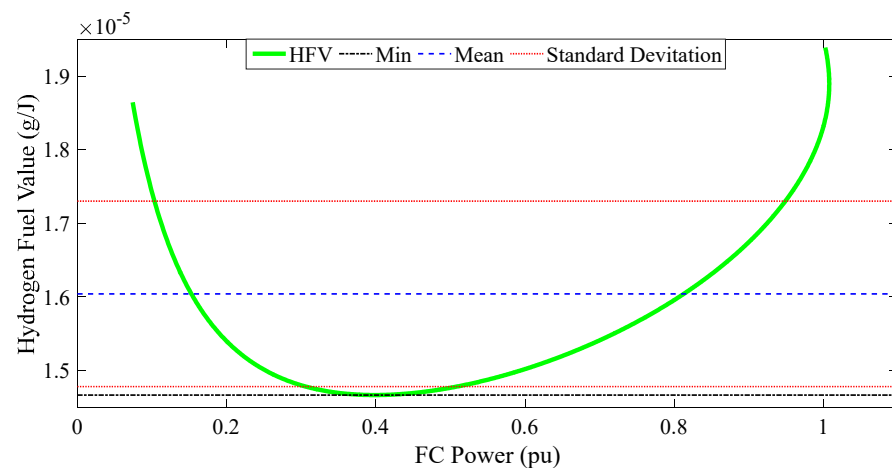


Figure 9. The total hydrogen value of the PEMFC system equipped with a linked step-up converter.

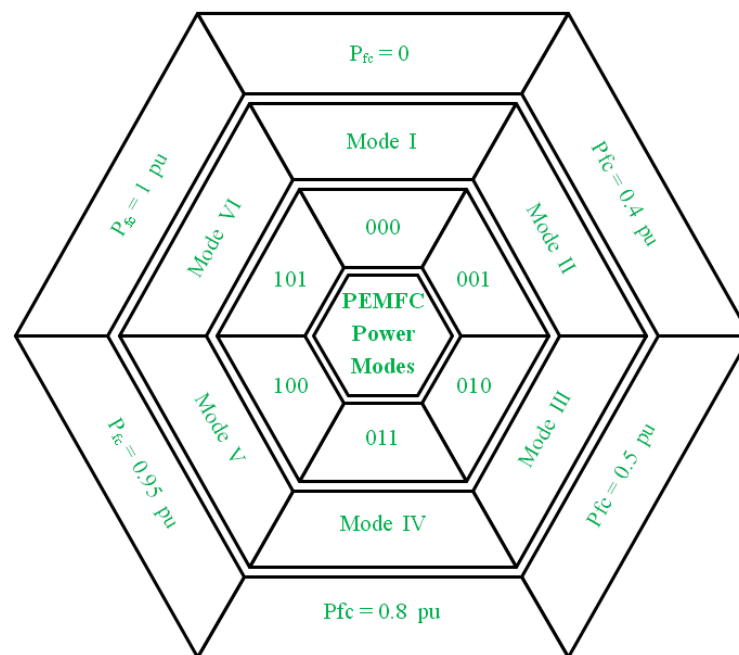


Figure 10. The binary codes of PEMFC in the proposed operational modes.

4.5. The Proposed Operational Mode Control Strategy

The proposed energy management strategy incorporates various states of robot movement, PEMFC production modes, P_{demand} modes, and the LFP battery SOC range as inputs. Based on the analysis of these inputs, the initial strategy has determined six operational modes for a fuel cell electric robot equipped with a LFP battery, specifying the type of PEMFC to be used. To determine the necessary power (P_{demand}), the fuel cell electric robot's various power modes are categorized as shown in Figure 11.

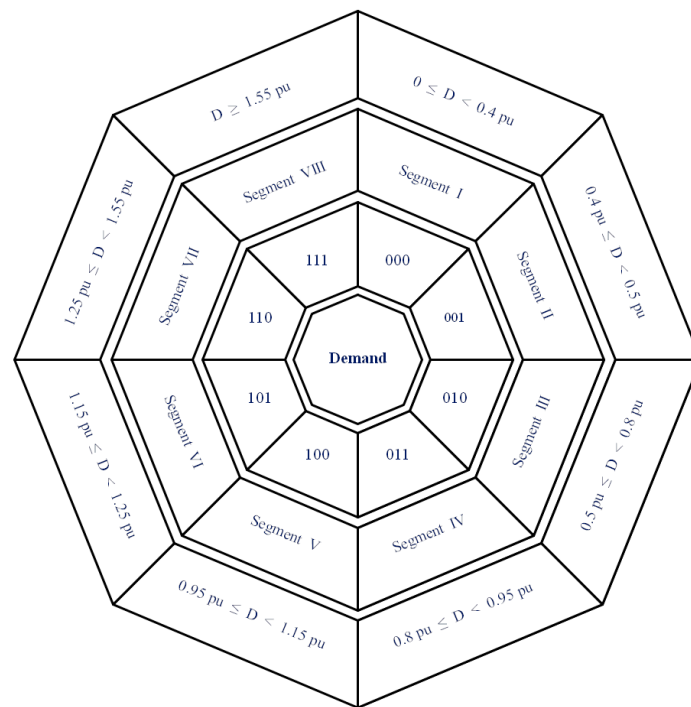


Figure 11. The binary codes of demand power in the proposed operational modes.

The proposed strategy for selecting PEMFC operational modes, which combines ECMS and SMC, has been formulated and calculated based on the following formulations:

- (1) The primary consideration when selecting an operational mode (i) is to obtain the optimal efficiency of the fuel cell with the lowest HCJE_{H_2} .
- (2) When the FC produces more power than the robot's demanded power (P_{demand}), the excess power charges the LFP battery. This is only allowed if the equivalent hydrogen consumption rate (HCJE_{H_2}) for the proposed mode is lower than the hydrogen consumption rate in the next battery operational mode, defined in Equation (27).

$$\text{LFP Charging Mode} = \left\{ i \mid \frac{(P_{\text{FC}}^{(i)} - \text{mean}(P_{\text{demand}})) \text{HCJE}_{\text{bat}}^{(i)}}{\text{mean}(P_{\text{demand}})} + \frac{P_{\text{FC}}^{(i)} \times \text{HCJE}_{\text{H}_2}^{(i)}}{\text{mean}(P_{\text{demand}})} \leq \text{HCJE}_{\text{H}_2}^{(i+1)} \right\} \quad (27)$$

- (3) Producing an FC power less than the demanded power (P_{demand}) results in the discharge of the LFP battery. This discharge is permitted only when the combined value of the equivalent total hydrogen consumption ($\text{HCJE}_{\text{bat}} + \text{HCJE}_{\text{H}_2}$) for the proposed mode is lower than the hydrogen rate. The consumption in the subsequent mode of the fuel cell is determined by Equation (28).

$$\text{LFP Discharging Mode} = \left\{ i \mid \frac{(\text{mean}(P_{\text{demand}}) - P_{\text{FC}}^{(i)}) \text{HCJE}_{\text{bat}}^{(i)}}{\text{mean}(P_{\text{demand}})} + \frac{P_{\text{FC}}^{(i)} \times \text{HCJE}_{\text{H}_2}^{(i)}}{\text{mean}(P_{\text{demand}})} \leq \text{HCJE}_{\text{H}_2}^{(i+1)} \right\} \quad (28)$$

- (4) Figure 12 calculates the recommended performance modes for different OSR modes and multiple demanded power state scenarios based on the given conditions. Figure 12 identifies 100 distinct scenarios for determining the proposed operational modes. Among the 20 available scenarios, the FC must operate on maximum power to be able to supply the demanded power. In two distinct scenarios, specifically those involving a high battery charge during the stationary state and traction state movement, the fuel cell is deactivated to minimize fuel consumption. Out of the remaining 78 scenarios, it is noted that the fuel cell operates at maximum overall efficiency with the lowest equivalent hydrogen consumption rate (PFC mode = 0.4 pu) in 46 scenarios, which

accounts for 59% of the remaining scenarios. The fuel cell electric robot's battery is charged in 7 different scenarios, and in each of these scenarios, the condition of Equation (27) is met. Additionally, the equivalent hydrogen consumption value is more optimal than the hydrogen consumption value of the subsequent mode. In the remaining 25 scenarios, the fuel cell electric robot's battery is discharged. In each of these scenarios, the condition of Equation (28) is met and the equivalent hydrogen consumption value is more efficient than the hydrogen consumption value of the subsequent mode.

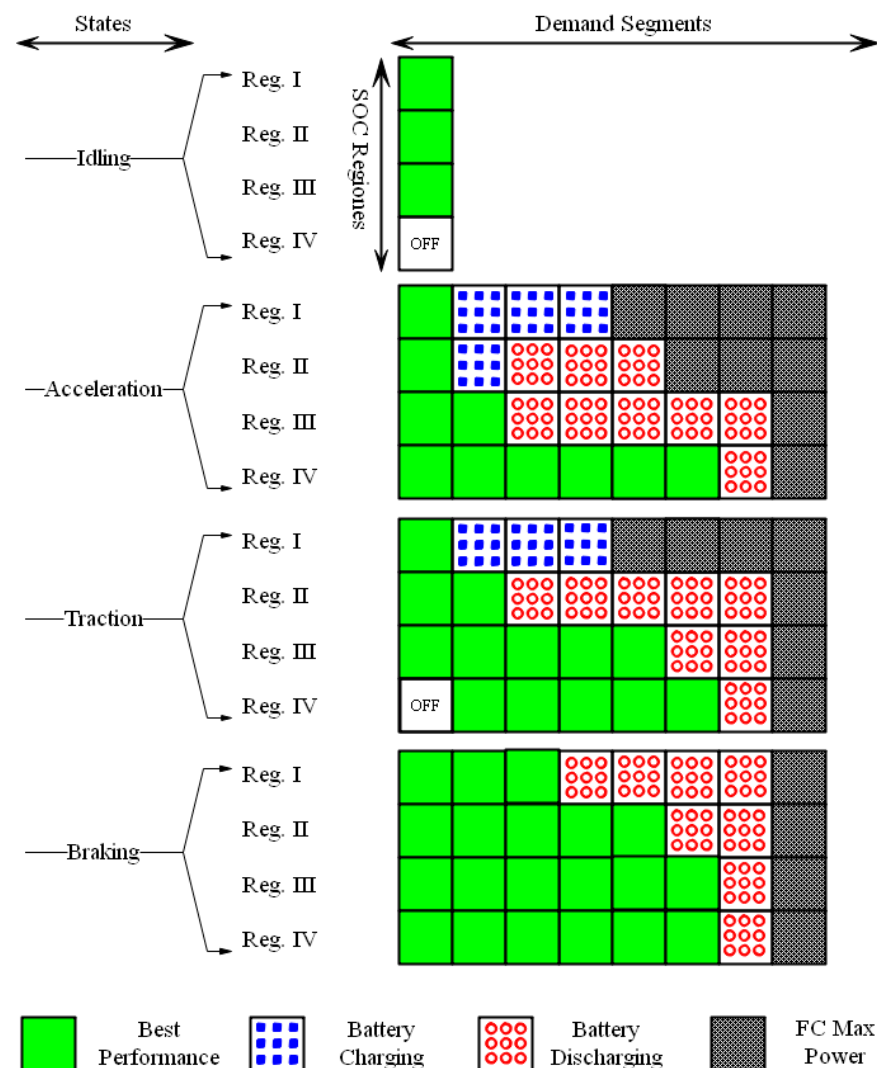


Figure 12. The static result of the proposed strategy for all possible scenarios in a fuel cell electric robot.

4.6. Battery Overcharge Control Strategy

The concept of variable battery overcharge control is introduced to reduce the disparity between the initial and final state of charge of the LFP battery. Based on the previously mentioned relations, the process of charging and discharging the battery incurs losses. Additionally, the current study has conducted calculations regarding the equivalent fuel consumption. Minimizing the difference between the initial and final state of charge (ΔSOC) is crucial in order to avoid energy loss in a fuel cell electric robot. Figure 12 shows that the battery of the fuel cell electric robot is charged in only 7 out of the 100 available scenarios. However, preventing the battery from being overcharged in the fuel cell electric robot can improve fuel consumption conditions.

If the battery's SOC is higher than its initial SOC and the battery can provide enough power on its own, the fuel cell is deactivated ($P_{FC} = 0$) to prevent fuel consumption. Once the battery's SOC drops below its initial level, the fuel cell will activate the necessary power supply circuit. Equation (29) expresses the dynamic power coefficient of the proposed variable.

$$BOCC(t) = u(SOC(t) - SOC_{init}) \times u(P_{demand}(t) - P_{B, rated}(t)) + u(SOC(t) - SOC_{init}) \times u(1 - |OSR(t)|) + u(SOC_{init} - SOC(t) + 0.001) \quad (29)$$

The inclusion of the second term in Equation (29) is necessary to ensure the battery is charged optimally when it is in a stationary state. Charging the battery under optimal conditions is most effective when the robot is in a stationary state ($OSR = 0$). Therefore, it is recommended to charge the battery under these ideal conditions. Additionally, the third term has been devised to mitigate the frequent connections/disconnections of the fuel cell caused by minor variations in battery SOC. This is performed to minimize the fluctuations in the fuel cell's output power within the proposed variable dynamic power factor formulation.

5. Simulation Results

The fuel cell electric robot's fuel cell system consists of a PEMFC stack with a power rating of 500 watts. It is equipped with an air-cooling system, and the specific parameters can be found in Table 1 [25]. The fuel cell has a minimum hydrogen consumption per joule of energy of 1.47×10^{-5} (g/joule) at $P_{FC} = 200$ watts, and a maximum consumption rate of 1.83×10^{-5} (g/joule) that occurs for maximum power generation.

Table 1. PEMFC parameters [25].

Items	Value	Unit
Rated Power	500	W
Rated Voltage	14.4	V
Rated Current	35	A
Cells Number	24	-
Stack Max Temperature	65	°C
Hydrogen Gas Pressure	0.45–0.55	bar

The $LiFePO_4$ battery, specifically the Ener POWER LFP battery cell model HTCFR18650, is used as an additional power source in the fuel cell electric robot. The detailed specifications of this battery can be found in Table 2 [26].

Table 2. LFP battery parameters [26].

Items	Value	Unit
Rated Power	5.76	W
Rated Voltage	3.2	V
Rated Capacity	1.8	Ah
Series Cells Number	8	-
Parallel Cells Number	3	-

Figure 13 displays the curve of the battery's equivalent hydrogen consumption ($HCJE_{bat}$) after simulating the equations mentioned in the previous section. Based on the figure, the minimum quantity of hydrogen consumed is observed when the state of charge is approximately 0.9 and the FC output power (P_{FC}) is around 0.4 per unit (pu).

Following the simulation of the fundamental equations of the fuel cell electric robot structure, an alternative P_{demand} curve was employed as a hypothetical input to verify the effectiveness of the proposed energy management strategy. The study applies the instantaneous power curve to the fuel cell electric robot structure without prior knowledge

and simulates it using MATLAB R2018b software. Additionally, it is assumed that the battery's initial charge status is 70%.

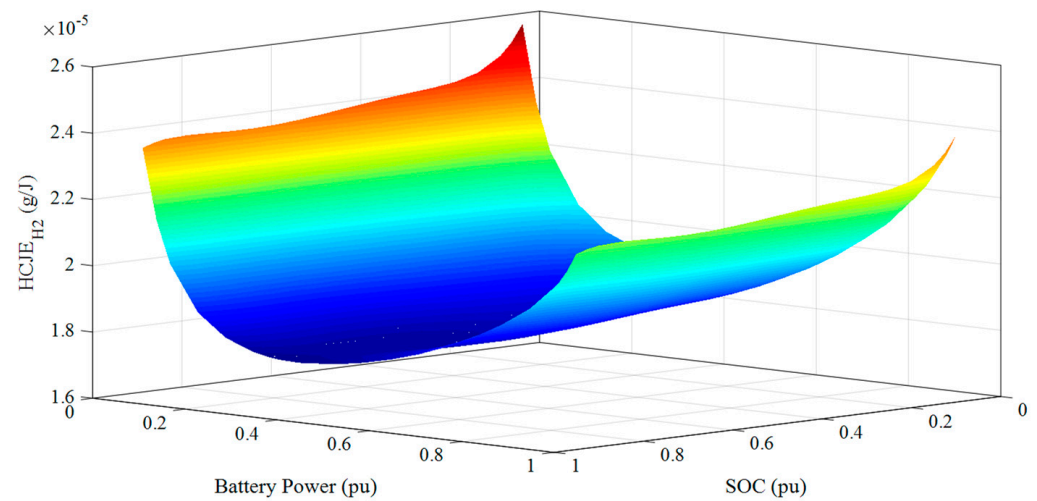


Figure 13. The LFP battery's equivalent hydrogen consumption per joule of energy curve.

Based on Figure 14, the fuel cell electric robot has applied the demanded power (P_{demand}) instantly for a duration of 135 s. During this period, the robot had no prior knowledge of the state of P_{demand} , allowing for an analysis of its online execution capability. Figures 15–18 display the simulation results for the energy management strategy and optimal power distribution of the fuel cell electric robot. The fuel cell electric robot's maneuver status identification function is computed in real time according to the equation in the previous section, and the results are displayed in Figure 15.

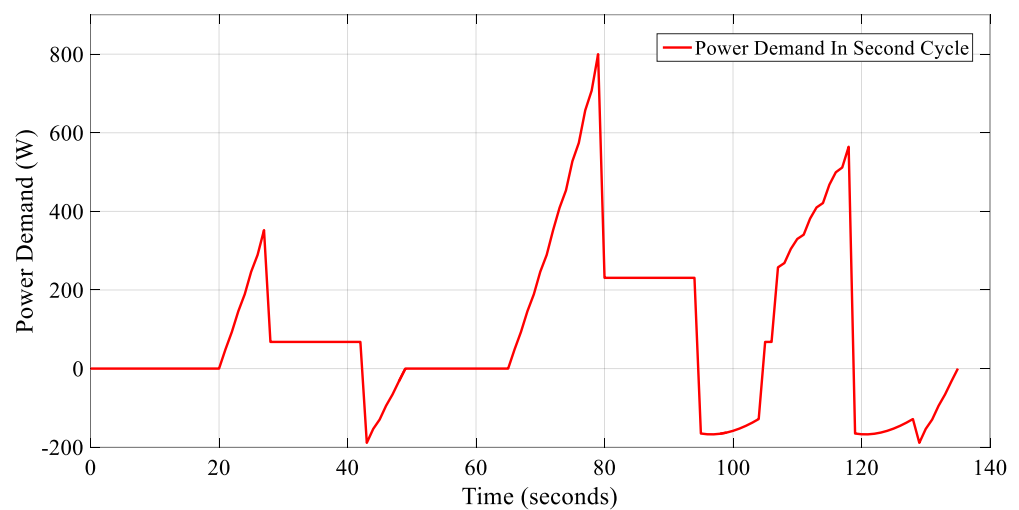


Figure 14. Demanded power pattern of the studied online cycle for a fuel cell electric robot.

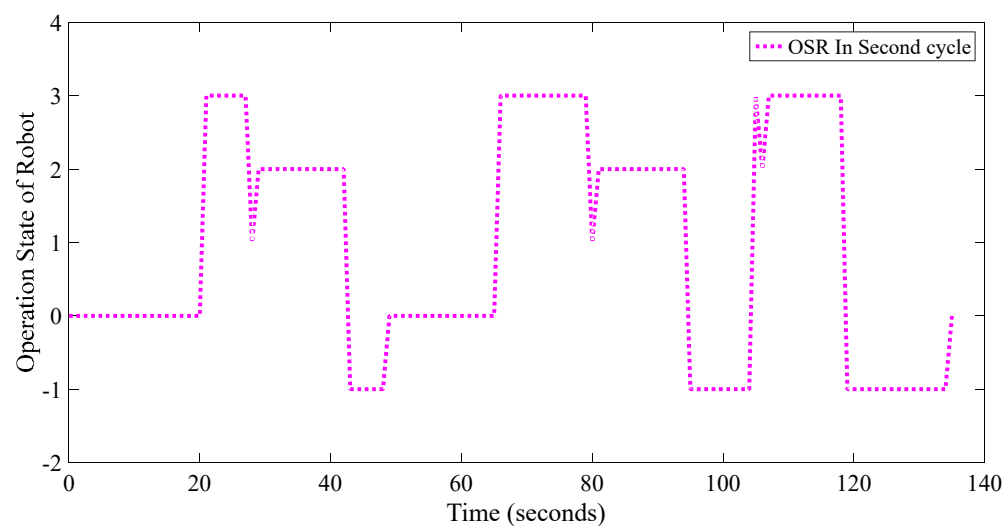


Figure 15. The maneuver state machine of the fuel cell electric robot under the studied online cycle.

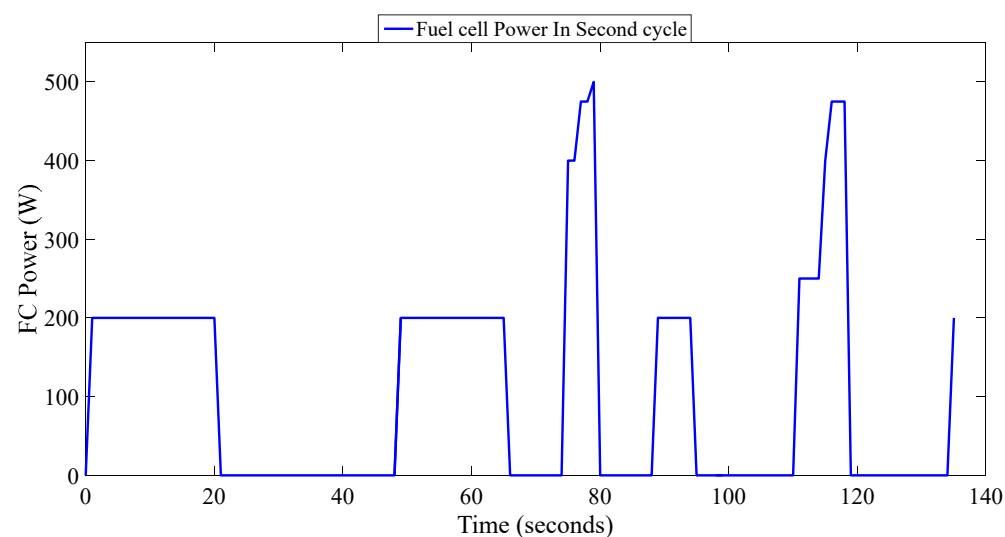


Figure 16. The PEMFC output power of the fuel cell electric robot under the studied online cycle.

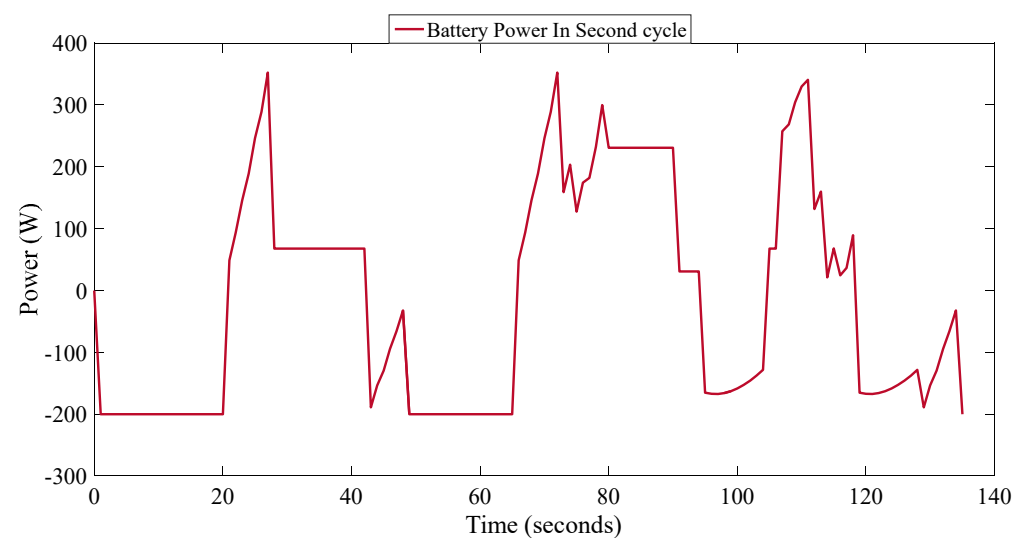


Figure 17. The battery output power of the fuel cell electric robot under the studied online cycle.

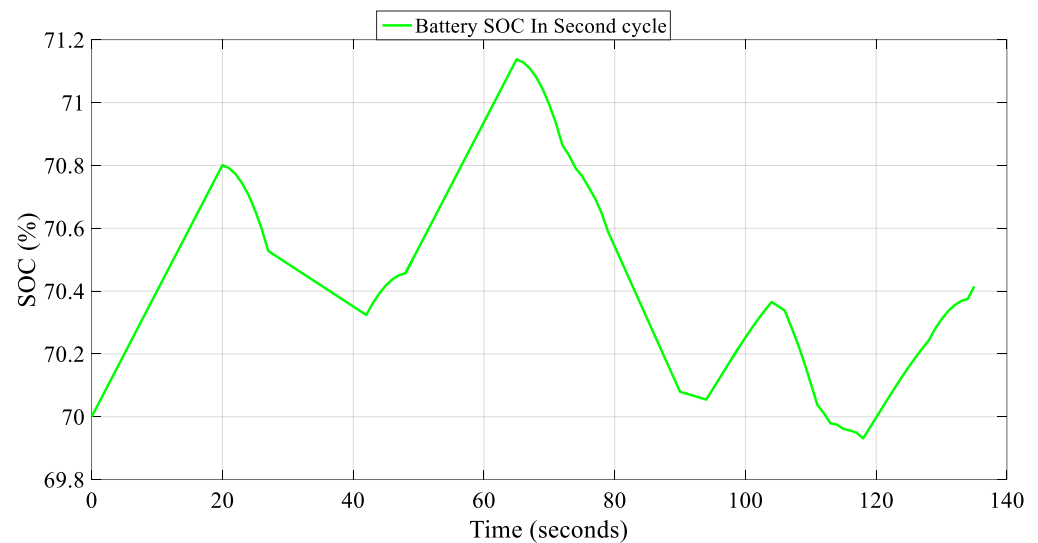


Figure 18. The LFP battery SOC of the fuel cell electric robot under the studied online cycle.

Figure 15 illustrates the maneuver state of the fuel cell electric robot's status function, which represents the ongoing changes in the robot's operational conditions. However, despite the continuous variations in P_{demand} and OSR, the strategy outlined in Figure 16 has effectively mitigated the fluctuations in the output power of the fuel cell. As depicted in the figure, while P_{demand} exhibits fluctuations, the fuel cell has undergone 15 changes in its output power status. Furthermore, the PEMFC has primarily been functioning at its peak efficiency point. Specifically, it has operated at maximum efficiency for a total of 43 s out of a 135-s period. Additionally, it has been inactive for 81 s and has only been active for 11 s at a power level other than the maximum point. The application of the proposed modes has successfully eliminated the PEMFC output power ripple, despite the sequential fluctuations of P_{demand} . In contrast, other energy management methods result in consecutive fluctuations of the fuel cell output power.

Figures 17 and 18 display the output power and SOC of the LFP battery. Table 3 indicates that the battery's maximum power for charging or discharging is 450.81 watts. Figure 17 demonstrates that battery charging only takes place when the fuel cell is operating at its most efficient level of energy consumption. This is supported by the relationships discussed in earlier sections, which show that this charging is justified as it helps to minimize fuel usage. Furthermore, it is important to mention that throughout this time-frame, the battery has successfully accumulated 4.4 kilojoules of energy generated through regenerative braking.

Table 3. Results' comparison.

EMS Strategies	Hybrid Rule Strategy via AMPSO [3]	Proposed Strategy
Hydrogen Consumption	0.484 g	0.214 g
Initial SOC	70%	70%
Final SOC	74.2%	70.4%
Δ SOC	+4.2%	+0.4%

Figure 18 displays the SOC of the LFP battery. The figure illustrates that the process of charging and discharging the battery is executed proficiently. It is evident that, even though 4.40 kilojoules of regenerative braking energy is received, the disparity between the initial and final battery charge status remains approximately 0.4%. The results demonstrate that the suggested approach effectively prevents the battery SOC from increasing. The SOC difference at the end of the robot's maneuver is consistently maintained at a positive level

(+0.4%) to ensure that the battery remains unaffected by various operating conditions and does not discharge.

Comparison of the Simulation Results

This study aims to verify the effectiveness of a power and energy management technique in a fuel cell electric robot that utilizes a fuel cell–battery system. The research compares the outcomes achieved with the incorporation of AMPSO optimization, as described in Reference [3]. The hybrid rule strategy via the AMPSO algorithm, as described in Reference [3], can be implemented in real time. Based on a thorough analysis of the theoretical sources, Reference [3] offers the most optimal solution available. Hence, comparing the outcomes of the suggested energy management approach with the hybrid rule strategy method using AMPSO will ensure the effectiveness and superiority of the suggested energy management strategy.

Figure 19 displays the output results of the Reference [3] for the power curve (P_{demand}). The findings from Figure 19, along with the comparison to Figures 15–18, suggest the following:

1. The method in [3] for the output power of PEMFC has been consistently utilized at various points, whereas the current research's proposed strategy exhibits fewer generation fluctuations and is only active at specific points.
2. The proposed method achieves an SOC difference of 0.4% ($\Delta\text{SOC} = 0.4\%$) at the end of the robot maneuver, while the method [3] yields an SOC difference of 4.02% ($\Delta\text{SOC} = 4.02\%$).
3. The hydrogen consumption for the power curve of the first study plan in Reference [3] is precisely 0.484 g. The proposed strategy involves the consumption of 0.214 g of hydrogen.

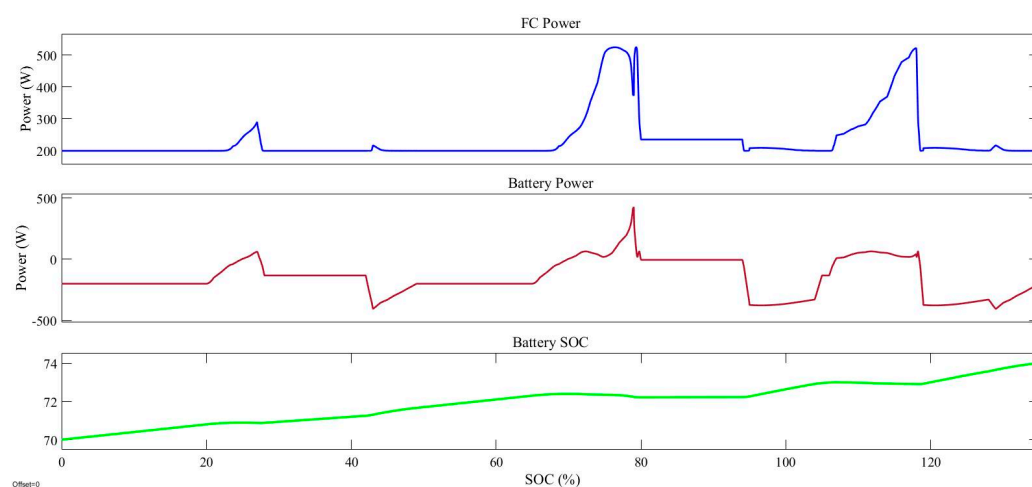


Figure 19. The output results of Reference [3] of the fuel cell electric robot under the studied online cycle.

The results obtained from the proposed energy management strategy are compared with the results of the prominent hybrid rule strategy via AMPSO [3] in Table 3. The hydrogen fuel consumption is measured in grams (g), while the change in the battery's state of charge (ΔSOC) is expressed as a percentage. Furthermore, the production power of PEMFC in the hybrid rule strategy using the AMPSO method [3] is not constrained to particular operating points and exhibits variations in fuel cell production power. In the proposed method, the output power of the PEMFC is maintained at the predetermined optimal operating points.

6. Conclusions

The provided materials outline the key findings of the proposed EMS, which incorporates state machine control, ECMS, operational mode control, and battery overcharge control. The summary of these results is as follows:

- (1) The OMC strategy has determined the operating conditions for the LFP battery to optimize its use. These conditions aim to minimize fluctuations in the output power of the PEMFC. Additionally, the strategy allows the battery to absorb energy from regenerative braking and deliver high power density when required.
- (2) In this research, a new formulation ($HCJE_{H_2}$) has been presented to solve the challenge of successive changes in the output power of the fuel cell (which leads to a decrease in the performance quality and lifespan of the fuel cell), which, after calculating the value of fuel consumption per joule of energy, the mathematical expectation and its standard deviation have been used to select the optimum points of fuel cell performance in order to reduce the successive fluctuations of the PEMFC generation power and to improve the performance of the ECMS and the operational mode control strategy.
- (3) Fuel cell operation at its global optimal point is a highly appealing operational concept in online energy management methods. However, maintaining the stability of the robot's activity is challenging due to its high level of mobility. In this study, precise monitoring of the conditions (OSR) and the differentiation of operating points for PEMFC, along with the OMC strategy, have been implemented. The objective is to track the global optimum point, which is achieved in more than 59% of cases (as shown in Figure 12), and to utilize a fuel cell operating at the point of maximum efficiency.
- (4) Presently, the issue of battery overcharging/discharging poses a significant obstacle in all online energy management strategies. This study introduces a novel approach called battery overcharge control (BOCC) that effectively regulates the change in the state of charge (ΔSOC) of the battery within an optimal range, ensuring a positive ($\Delta SOC > 0$) and preventing the overcharging/discharging of the battery. The implementation of the new function, known as BOCC, has successfully achieved a decrease in hydrogen fuel consumption by effectively preventing the battery from undergoing unnecessary charging and discharging. Occasionally, the output results from previous references have shown a negative range of ΔSOC ($\Delta SOC < 0$), indicating inadequate long-term management and complete discharge of the battery over time.
- (5) The results obtained suggest that, primarily, the PEMFC fuel cell consistently operated at its highest level of efficiency, without any fluctuations in power output, in the majority of cases. Additionally, the variation in the state of charge (ΔSOC) of the battery was found to be less than 0.4% across various research protocols. The aforementioned findings demonstrate the superiority of the proposed approach in enhancing energy efficiency, diminishing fuel usage, and prolonging the lifespan of hybrid power sources, such as batteries and fuel cells.
- (6) The simulation results of the comparison between the proposed strategy and other approaches demonstrate a decrease in hydrogen consumption and the prevention of battery overcharging.

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Abbreviations

The following abbreviations are used in this manuscript:

BCE	Battery combined efficiency
BOCC	Battery overcharge control
EHCFV	Equivalent hydrogen consumption fuel value
ECM	Equivalent consumption minimization
ECMS	Equivalent consumption minimization strategy
EMS	Energy management strategy
FC	Fuel cell
LFP	LiFePO ₄ battery
OMCS	Operational mode control strategy
OSR	The operational state of the robot
PEMFC	Proton exchange membrane fuel cell
PDF	Probability density function
SBCE	Square of battery combined efficiency
SMC	State machine control
SMCS	State machine control strategy
SOC	State of charge
UAV	Unmanned aerial vehicle
pu	Per unit

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